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United States Patent	12404962
Kind Code	B2
Date of Patent	September 02, 2025
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Connector assembly

Abstract

A connector assembly is provided. The connector assembly includes a male connector for connecting with a female connector. The female connector includes: a housing; a first tubular body received within and fixed to the housing; and a locking sleeve rotatably sleeved outside the first tubular body and including a guiding groove and a locking groove. The guiding groove and the locking groove are arranged on a circumferential wall of the locking sleeve and are connected to each other. The male connector includes: a casing; and a second tubular body received within and fixed to the casing and including a locking protrusion arranged on an outer periphery thereof. The connector assembly is configured to realize the connecting and locking of the female connector and the male connector in a small operation space.

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Appl. No.:	18/294538
Filed (or PCT Filed):	August 01, 2022
PCT No.:	PCT/CN2022/109411
PCT Pub. No.:	WO2023/011406
PCT Pub. Date:	February 09, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240328552 A1	Oct. 03, 2024

Foreign Application Priority Data

Publication Classification

Int. Cl.: F16L37/34 (20060101)

U.S. Cl.:

CPC F16L37/34 (20130101);

Field of Classification Search

CPC: F16L (37/34); F16L (37/35); Y10T (137/87957); Y10T (137/87973)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the National Stage of International Appl. No. PCT/CN2022/109411 filed on 1 Aug. 2022, which claims priority to and all advantages of Chinese Patent Appl. No. 202110881106.7 filed on 2 Aug. 2021 and Chinese Patent Appl. No. 202121780013.7 filed on 2 Aug. 2021, the contents of which are hereby incorporated by reference.

RELATED FIELD

(2) The present application generally relates to a connector assembly for establishing fluid communication.

BACKGROUND

(3) A connector assembly can be used to establish fluid communication between fluid pipelines in many different application scenarios.

(4) The connector assembly generally includes a female connector and a male connector each of which is directly or indirectly connected to a fluid pipeline. The male connector can be inserted into the female connector to establish fluid communication between the fluid pipelines. In some application scenarios, the connector assembly is required to have a bidirectional cut-off function to prevent the fluid in the fluid pipelines from flowing out through the female connector and the male connector when the female connector and the male connector are disconnected. In some other application scenarios, such as in the case that the connector assembly is applied to a battery pack of a new energy vehicle, it is desirable to quickly and conveniently connect the female connector with the male connector in a small installation space. In addition, the connector assembly with the bidirectional cut-off function is needed in many application scenarios, so it is desirable to improve the compatibility of the connector assembly for various application scenarios. At present, developing a connector assembly that can achieve the above functions still faces many challenges.

SUMMARY

(5) An object of the present application is to provide an improved connector assembly to achieve one or more of the above functions.

(6) The present application provides a connector assembly which includes a female connector and a male connector for being connected with the female connector. The female connector includes: a housing; a first tubular body defining a first axial direction, the first tubular body being at least partially received within and being fixed to the housing; a valve unit at least partially arranged in the first tubular body; and a locking sleeve sleeved outside the first tubular body in a rotatable manner. At least part of the locking sleeve is confined between the housing and the first tubular body in the first axial direction, and the locking sleeve includes a guiding groove and a locking groove which are connected to each other and arranged on a circumferential wall of the locking sleeve. The male connector includes: a casing; a second tubular body defining a second axial direction, the second tubular body being at least partially received within and being fixed to the casing, the second tubular body being configured to be insertable between the locking sleeve and the first tubular body in the first axial direction, the second tubular body including a locking protrusion arranged on an outer periphery of the second tubular body; and a valve assembly at least partially arranged in the second tubular body. During an insertion of the second tubular body between the locking sleeve and the first tubular body, the locking protrusion is adapted to move along the guiding groove and drive the locking sleeve to rotate until the locking protrusion enters the locking groove. The locking protrusion is adapted to engage with the locking groove due to interaction of the valve assembly and the valve unit after the locking protrusion enters the locking

groove, to prevent the second tubular body from disengaging from the locking sleeve.

(7) Arranging the locking sleeve in a rotatable manner allows the female connector to be connected and locked with the male connector by means of straight insertion/push operation of the female connector and/or male connector during the connection of the female connector and the male connector. This straight insertion/push operation requires a smaller operation space and is convenient to conduct. And during the straight insertion/push operation, there is no need to rotate the housing of the female connector. This avoids the rotational resistance due to the fluid pipeline connected to the housing of the female connector when the housing of the female connector is rotated, and also reduces stress at the connection between the housing of the female connector and the fluid pipeline connected thereto.

(8) According to the above technical concept, the present application may further include one or more of the following optional forms.

(9) In some optional forms, the housing has a first limiting portion. The first tubular body has a second limiting portion arranged on the outer periphery of the first tubular body. The locking sleeve has a shoulder extending radially inward. And the shoulder is confined between the first limiting portion and the second limiting portion.

(10) In some optional forms, the locking sleeve has a first end and a second end. And the guiding groove extends from the first end towards the second end, and the guiding groove has an inlet end at the first end and an outlet end away from the first end.

(11) In some optional forms, the guiding groove is arc-shaped.

(12) In some optional forms, an angle between a first radial direction of the locking sleeve pointed towards the inlet end and a second radial direction of the locking sleeve pointed towards the outlet end is less than 90° .

(13) In some optional forms, the angle ranges from 30° to 60° .

(14) In some optional forms, the angle is 45° .

(15) In some optional forms, the locking groove extends in the first axial direction, and at least part of the locking groove extends from the outlet end of the guiding groove towards the first end of the locking sleeve.

(16) In some optional forms, the locking groove has a positioning end and a locking end opposite to each other. The positioning end and the locking end are respectively located on two sides of the outlet end of the guiding groove. The locking end is closer to the first end of the locking sleeve than the positioning end. The locking protrusion is adapted to move along the guiding groove and enter the positioning end during the insertion of the second tubular body between the locking sleeve and the first tubular body. And the locking protrusion is further adapted to move from the positioning end to the locking end and engage with the locking end due to the interaction of the valve assembly and the valve unit after the locking protrusion enters the positioning end.

(17) In some optional forms, the housing includes a first receiving section and a second receiving section which are configured to be capable of being mated with the first tubular body and the second tubular body, respectively; and the casing includes a third receiving section and a fourth receiving section which are configured to be capable of being mated with the first tubular body and the second tubular body, respectively. In this way, the connector assembly with various configurations can be provided and the application range of the connector assembly can be expanded, while the manufacturing cost of the connector assembly, especially the cost of preparing the mold can be significantly reduced.

(18) In some optional forms, the housing includes a first connection section in the form of a hose connection section or a threaded connection section; and/or the casing includes a second connection section in the form of a hose connection section or a threaded connection section.

(19) In some optional forms, the housing and the first tubular body are fixed to each other by interference fit, snap fit and/or welding; and/or the casing and the second tubular body are fixed to each other by interference fit, snap fit and/or welding.

- (20) In some optional forms, the first tubular body has a plug end defining a port. The valve unit includes a valve core and an elastic member. The valve core is movable in the first axial direction between a first closed position and a first open position. The valve core is biased toward the first closed position by the elastic member. The valve core blocks the port to cut off a flow path of the female connector when the valve core is in the first closed position, and the flow path of the female connector is opened when the valve core is in the first open position.
- (21) In some optional forms, the plug end defines a first inner peripheral inclined surface and a second inner peripheral inclined surface. An inner peripheral surface of the port, the first inner peripheral inclined surface, and the second inner peripheral inclined surface are sequentially connected in the first axial direction. The first inner peripheral inclined surface and the second inner peripheral inclined surface are configured to guide a sealing member embedded in an outer periphery of the valve core to move in the first axial direction to come into sealing contact with the inner peripheral surface of the port. The first inner peripheral inclined surface and the second inner peripheral inclined surface respectively form a first angle and a second angle with respect to the first axial direction, and the first angle is smaller than the second angle.
- (22) In some optional forms, the first angle ranges from 5° to 15° .
- (23) In some optional forms, the valve assembly includes a valve stem, a sliding sleeve and an elastic element. The valve stem includes a valve stem head portion and a valve stem base portion at two ends of the valve stem, respectively. The sliding sleeve is arranged in the second tubular body and sleeved outside the valve stem, and the sliding sleeve is slidable between a second closed position and a second open position along the second axial direction. And two ends of the elastic element respectively abut against the sliding sleeve and the valve stem base portion to bias the sliding sleeve toward the second closed position. The sliding sleeve blocks an annular gap between the second tubular body and the valve stem head portion to cut off a flow path of the male connector when the sliding sleeve is in the second closed position, and the flow path of the male connector is opened when the sliding sleeve is in the second open position.
- (24) In some optional forms, the valve stem base portion includes: a base body provided with a through hole penetrating through the base body for fluid to flow therethrough; and a projection extending from a center of the base body and away from the valve stem head portion. The projection tapers in a direction away from the valve stem head portion.
- (25) In some optional forms, the projection is in the shape of a semi-ellipsoid.
- (26) In some optional forms, the second tubular body includes a second tubular main body. The locking protrusion is made of metal, and the second tubular main body is made of plastic. And the second tubular main body and the locking protrusion are integrally formed by insert injection molding.
- (27) The connector assembly according to the present application can conveniently realize the connecting and locking of the female connector and the male connector in a small operation space, and the configuration of the connector assembly can be varied at a low cost to adapt to various application scenarios.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other features and advantages of the present application will be readily understood through the following preferred embodiments described in detail with reference to the accompanying drawings, in which the same reference numerals indicate the same or similar components.
- (2) FIG. 1 is a perspective view of a connector assembly according to a first embodiment of the present application;
- (3) FIG. 2A and FIG. 2B respectively are a perspective view and a front view of a female connector

of the connector assembly according to the first embodiment of the present application;

(4) FIG. 3A is a cross-sectional view of the female connector of the connector assembly according to the first embodiment of the present application;

(5) FIG. 3B is a partly enlarged view of FIG. 3A;

(6) FIG. 4A to FIG. 4C respectively are a perspective view, a front view, and a cross-sectional view of a housing of the female connector of the connector assembly according to the first embodiment of the present application;

(7) FIG. 5A and FIG. 5B respectively are a perspective view and a front view of a first tubular body of the female connector of the connector assembly according to the first embodiment of the present application;

(8) FIG. 6A and FIG. 6B respectively are perspective views of a locking sleeve of the female connector of the connector assembly according to the first embodiment of the present application, viewed from different perspectives;

(9) FIG. 6C is a front view of the locking sleeve of the female connector of the connector assembly according to the first embodiment of the present application;

(10) FIG. 7A and FIG. 7B respectively are a perspective view and a front view of a valve core of the female connector of the connector assembly according to the first embodiment of the present application;

(11) FIG. 8A and FIG. 8B respectively are a perspective view and a front view of a male connector of the connector assembly according to the first embodiment of the present application;

(12) FIG. 9 is a cross-sectional view of the male connector of the connector assembly according to the first embodiment of the present application;

(13) FIG. 10A and FIG. 10B respectively are perspective views of a casing of the male connector of the connector assembly according to the first embodiment of the present application, viewed from different perspectives;

(14) FIG. 10C and FIG. 10D respectively are a front view and a cross-sectional view of the casing of the male connector of the connector assembly according to the first embodiment of the present application;

(15) FIG. 11A and FIG. 11B respectively are a perspective view and a front view of a second tubular body of the male connector of the connector assembly according to the first embodiment of the present application;

(16) FIG. 12A and FIG. 12B respectively are a perspective view and a front view of a valve stem of the male connector of the connector assembly according to the first embodiment of the present application;

(17) FIG. 13A and FIG. 13B respectively are a perspective view and a front view of a sliding sleeve of the male connector of the connector assembly according to the first embodiment of the present application;

(18) FIG. 14A to FIG. 14C respectively are a perspective view, a front view, and a cross-sectional view of the female connector and the male connector of the connector assembly according to the first embodiment of the present application, when the female connector and the male connector are disconnected from each other;

(19) FIG. 15A to FIG. 15C respectively are a perspective view, a front view, and a cross-sectional view of the female connector and the male connector of the connector assembly according to the first embodiment of the present application, when the female connector and the male connector are in initial contact with each other during the connection thereof;

(20) FIG. 16A to FIG. 16C respectively are a perspective view, a front view, and a cross-sectional view of the female connector and the male connector of the connector assembly according to the first embodiment of the present application, when a locking protrusion is moving along a guiding groove during the connection of the female connector and the male connector;

(21) FIG. 17A to FIG. 17C respectively are a perspective view, a front view, and a cross-sectional

view of the female connector and the male connector of the connector assembly according to the first embodiment of the present application, when the locking protrusion enters a positioning end of a locking groove during the connection of the female connector and the male connector;

(22) FIG. **18A** to FIG. **18C** respectively are a perspective view, a front view, and a cross-sectional view of the female connector and the male connector of the connector assembly according to the first embodiment of the present application, when the female connector and the male connector are locked to each other, wherein the locking protrusion is engaged with a locking end of the locking groove;

(23) FIG. **19A** to FIG. **19C** respectively are a perspective view, a front view, and a cross-sectional view of a connector assembly according to a second embodiment of the present application;

(24) FIG. **20A** to FIG. **20C** respectively are a perspective view, a front view, and a cross-sectional view of a connector assembly according to a third embodiment of the present application;

(25) FIG. **21A** to FIG. **21C** respectively are a perspective view, a front view, and a cross-sectional view of a connector assembly according to a fourth embodiment of the present application; and

(26) FIG. **22A** to FIG. **22C** respectively are a perspective view, a front view, and a cross-sectional view of a connector assembly according to a fifth embodiment of the present application.

DETAILED DESCRIPTION

(27) The implementation and usage of the embodiments are discussed in detail below. However, it is conceivable that the specific embodiments discussed are merely intended to illustrate specific ways of implementing and using the present application, and are not intended to limit the scope of the present application. When describing structures and positions of components, the direction-related expressions herein, such as “upper”, “lower”, “top”, and “bottom”, are not absolute, but relative. When the components are arranged as shown in the drawings, these direction-related expressions are appropriate, but when the positions of these components in the drawings are altered, these direction-related expressions should be altered accordingly.

(28) In the present application, an axial direction of a tubular or annular component refers to a direction along the central axis of the component, a circumferential direction of the tubular or annular component refers to a direction along the circumference of the component, and a radial direction of the tubular or annular component refers to a direction passing through the central axis of the component and perpendicular to the axial direction of the component.

(29) FIG. **1** to FIG. **18C** illustrate a connector assembly **10** according to a first embodiment of the present application.

(30) Referring to FIGS. **1** to **3B** and FIGS. **8A** to **9**, the connector assembly **10** includes a female connector **100** and a male connector **200** for being connected with the female connector **100**. The female connector **100** includes a housing **102**, a first tubular body **104**, a valve unit **106** and a locking sleeve **108**. The first tubular body **104** defines a first axial direction **A1**. The first tubular body **104** is at least partially received within the housing **102** and is fixed to the housing **102**. The valve unit **106** is at least partially arranged in the first tubular body **104**. The locking sleeve **108** is sleeved outside the first tubular body **104** in a rotatable manner. At least part of the locking sleeve **108** is confined between the housing **102** and the first tubular body **104** in the first axial direction **A1**. The locking sleeve **108** includes a guiding groove **110** and a locking groove **112** which are connected to each other and arranged on a circumferential wall of the locking sleeve **108**. The male connector **200** includes a casing **202**, a second tubular body **204** and a valve assembly **206**. The second tubular body **204** defines a second axial direction **A2**. The second tubular body **204** is at least partially received within the casing **202** and is fixed to the casing **202**. The second tubular body **204** is configured to be insertable between the locking sleeve **108** and the first tubular body **104** in the first axial direction **A1**. The second tubular body **204** includes a locking protrusion **208** arranged on an outer periphery of the second tubular body **204**. The valve assembly **206** is at least partially arranged in the second tubular body **204**. While the second tubular body **204** is inserted between the locking sleeve **108** and the first tubular body **104**, the locking protrusion **208** is

adapted to move along the guiding groove **110** and drive the locking sleeve **108** to rotate until the locking protrusion **208** enters the locking groove **112**. The locking protrusion **208** is adapted to engage with the locking groove **112** due to the interaction of the valve assembly **206** and the valve unit **106** after the locking protrusion **208** enters the locking groove **112**, to prevent the second tubular body **204** from disengaging from the locking sleeve **108**.

(31) FIGS. 2A to 7B illustrate the female connector **100** of the connector assembly **10** according to the first embodiment of the present application and its various components. The housing **102**, the first tubular body **104** and the locking sleeve **108** of the female connector **100** may all be formed by injection molding.

(32) Referring to FIG. 3A and FIGS. 4A to 4C, the housing **102** of the female connector **100** may be substantially tubular, and include a first connection section **114**, a first receiving section **116** and a second receiving section **118** connected to each other in the first axial direction **A1**. The first connection section **114** may be directly connected to a fluid pipeline for fluid communication with the fluid pipeline. In the illustrated embodiment, the first connection section **114** may be in the form of a hose connection section to be directly connected with a hose (not shown). The first connection section **114** includes multiple circumferential annular flanges **120** on the outer periphery of the first connection section **114**, and the hose can be sleeved outside the first connection section **114** and remain connected to the housing **102** through the multiple annular flanges **120**.

(33) Referring to FIG. 5A and FIG. 5B, the first tubular body **104** of the female connector **100** is substantially tubular, and includes a fixing end **122** and a plug end **124** in the first axial direction **A1**. The plug end **124** of the first tubular body **104** defines a port **126** (as shown in FIG. 18C).

(34) Referring to FIG. 3A and FIG. 4C, in the illustrated embodiment, the first tubular body **104** of the female connector **100** is at least partially received within the first receiving section **116** of the housing **102** and is fixed to the first receiving section **116** of the housing **102**.

(35) The first receiving section **116** of the housing **102** is configured to be capable of being mated with the fixing end **122** of the first tubular body **104**. Optionally, the first receiving section **116** may be in interference fit with the fixing end **122** of the first tubular body **104**, such that the fixing end **122** of the first tubular body **104** is fixed to the first receiving section **116**.

(36) Optionally, the first receiving section **116** and the fixing end **122** of the first tubular body **104** may be further fixed to each other by welding in the circumferential direction to provide higher fixing strength and achieve the sealing between the first receiving section **116** and the first tubular body **104**, thereby reducing the use of sealing members and saving costs.

(37) Optionally, as shown in FIG. 3A and FIG. 4C, the outer peripheral surface of the fixing end **122** of the first tubular body **104** may be provided with a first positioning protrusion **128**, the inner peripheral surface of the first receiving section **116** may correspondingly be provided with a first positioning recess **130**, and the first positioning protrusion **128** and the first positioning recess **130** may be engaged with each other by snap fit, so as to improve the fixing strength of the first tubular body **104** and the first receiving section **116**.

(38) It is conceivable that the housing **102** and the first tubular body **104** may be fixed to each other by interference fit, snap fit, welding or any combination thereof.

(39) In addition, referring to FIG. 3A, FIG. 4C and FIG. 14C, the second receiving section **118** of the housing **102** is configured to be capable of being mated with the second tubular body **204** of the male connector **200**. In other words, according to the actual application requirements, for example, in the case that the casing of the male connector **200** needs to have a hose connection section, the housing **102** of the female connector **100** can also be used as the casing of the male connector **200** and be assembled and fixed with the second tubular body **204** of the male connector **200**, so that there is no need to manufacture a new mold for a casing having a hose connection section for the male connector **200**.

(40) Thus, the first receiving section **116** and the second receiving section **118** of the housing **102** can respectively be mated with the first tubular body **104** of the female connector **100** and the

second tubular body **204** of the male connector **200**, which can reduce the number of injection molds required for manufacturing the connector assembly **10** and significantly reduce manufacturing costs, as will be explained further below.

(41) The first receiving section **116** has a larger diameter than the first connection section **114** and the second receiving section **118** has a larger diameter than the first receiving section **116**, so that a first inner stepped portion **132** is formed between the first connection section **114** and the first receiving section **116**, and a second inner stepped portion **134** is formed between the first receiving section **116** and the second receiving section **118**. The first inner stepped portion **132** is configured to limit the first tubular body **104** when the first receiving section **116** is mated with the first tubular body **104**.

(42) Referring to FIGS. 3A, 3B, 7A and 7B, in the illustrated embodiment, the valve unit **106** may be arranged within the first tubular body **104**. The valve unit **106** may include a valve core **136** and an elastic member **138**. The valve core **136** is movable in the first axial direction **A1** between a first closed position (shown in FIG. 3B) and a first open position (shown in FIG. 18C). The valve core **136** is biased toward the first closed position by the elastic member **138**. The valve core **136** blocks the port **126** to cut off a flow path of the female connector **100** when the valve core **136** is in the first closed position, and the flow path of the female connector **100** is opened when the valve core **136** is in the first open position.

(43) The valve core **136** may include a valve core head **140** and a bracket **142**. A sealing member **144** may be provided between the outer peripheral surface of the valve core head **140** and the inner peripheral surface of the first tubular body **104**. In the illustrated embodiment, the sealing member **144** is embedded in the outer periphery of the valve core head **140** for sealing contact with the inner peripheral surface of the port **126** of the plug end **124**. The valve core **136** may be formed by injection molding.

(44) One end of the elastic member **138** may abut against the bracket **142** of the valve core **136**, and the other end of the elastic member **138** may abut against the first inner stepped portion **132** of the housing **102**, so as to bias the valve core **136** toward the first closed position for blocking the port **126**. When the valve core **136** is biased in the first closed position by the elastic member **138**, the valve core head **140** is in sealing contact with the inner peripheral surface of the port **126**, so that the flow path of the female connector **100** is cut off. When the valve core **136** is pushed by an external force along the first axial direction **A1**, the valve core **136** can move away from the port **126** to the first open position against the elastic force of the elastic member **138**, so that the flow path of the female connector **100** is opened. The elastic member **138** may be in the form of a coil spring.

(45) As shown in FIG. 18C, when the valve core **136** is in the first open position, fluid can enter the first tubular body **104** from the port **126** of the first tubular body **104** and flow through a gap between the first tubular body **104** and the valve core **136**, and then flow into the fluid pipeline (not shown) connected with the female connector **100**. And vice versa, fluid may flow along a reverse direction into the first tubular body **104** from the fluid pipeline (not shown) connected with the female connector **100** and then flow out of the port **126** of the first tubular body **104**.

(46) Referring to FIG. 3B again, the inner periphery of the plug end **124** may define a first inner peripheral inclined surface **146** and a second inner peripheral inclined surface **148**. The inner peripheral surface of the port **126**, the first inner peripheral inclined surface **146**, and the second inner peripheral inclined surface **148** are sequentially connected in the first axial direction **A1**. The first inner peripheral inclined surface **146** and the second inner peripheral inclined surface **148** are configured to guide the sealing member **144** embedded in the outer periphery of the valve core **136** to move in the first axial direction **A1** into sealing contact with the inner peripheral surface of the port **126**. The first inner peripheral inclined surface **146** and the second inner peripheral inclined surface **148** respectively form a first angle and a second angle with respect to the first axial direction **A1**, and the first angle is smaller than the second angle. The first angle may range from,

for example, 5° to 15°. By guiding the sealing member **144** of the valve core **136** into the port **126** along the two inner peripheral inclined surfaces, inclination angles of which with respect to the first axial direction **A1** decrease in turn, the elastic force, to be provided by the elastic member **138** and required for the valve core **136** to move, for example, from the first open position to the first closed position for blocking the port **126**, may be reduced. Since the elastic member **138** only needs to provide a small elastic force, the service life of the elastic member **138** can be prolonged, thereby improving the service life of the female connector **100**. In addition, this arrangement allows using an elastic member **138** with a lower elastic modulus for returning the valve core **136** to the first closed position, making assembly of the female connector **100** easier.

(47) Referring to FIGS. **3A**, **4C**, **5A**, and **6A** to **6C**, the locking sleeve **108** of the female connector **100** may be coaxially sleeved outside the first tubular body **104** in a rotatable manner.

(48) The locking sleeve **108** has a first end **150** and a second end **152**. The locking sleeve **108** may further have a shoulder **154** extending radially inward. The housing **102** may have a first limiting portion **156**. The first tubular body **104** may have a second limiting portion **158** arranged on the outer periphery of the first tubular body **104**. The shoulder **154** of the locking sleeve **108** may be confined between the first limiting portion **156** and the second limiting portion **158**. In the illustrated embodiment, the first limiting portion **156** of the housing **102** may be an axial end of the housing **102**, the second limiting portion **158** of the first tubular body **104** may be in the form of an annular flange, and the shoulder **154** of the locking sleeve **108** may be annular and located at the second end **152** of the locking sleeve **108**.

(49) The locking sleeve **108** may include the guiding groove **110** and the locking groove **112** which penetrate through the circumferential wall of the locking sleeve **108** and are connected to each other. In the illustrated embodiment, the guiding groove **110** may extend from the first end **150** toward the second end **152** and have an arc shape. The guiding groove **110** may have an inlet end **160** at the first end **150** of the locking sleeve **108** and an outlet end **162** away from the first end **150**.

(50) The locking groove **112** extends in the first axial direction **A1**, and at least part of the locking groove **112** extends from the outlet end **162** of the guiding groove **110** towards the first end **150** of the locking sleeve **108**. In the illustrated embodiment, the locking groove **112** has a positioning end **164** and a locking end **166** opposite to each other. The positioning end **164** and the locking end **166** are respectively located on two sides of the outlet end **162** of the guiding groove **110**. The locking end **166** is closer to the first end **150** of the locking sleeve **108** than the positioning end **164**.

(51) Referring to FIGS. **14A** to **18C**, while the second tubular body **204** of the male connector **200** is inserted between the locking sleeve **108** and the first tubular body **104** of the female connector **100** under the action of an external force, the locking protrusion **208** moves along the guiding groove **110** and enters the positioning end **164**. After the locking protrusion **208** enters the positioning end **164** of the locking groove **112** and the external force is removed, the locking protrusion **208** would move from the positioning end **164** to the locking end **166** and engage with the locking end **166** to lock the female connector **100** to the male connector **200** since the valve assembly **206** and the valve unit **106** interact with each other and tend to disengage from each other. This will be explained in detail below.

(52) With reference to FIGS. **6B** and **6C**, an angle between a first radial direction **R1**, pointed towards the inlet end **160**, of the locking sleeve **108** and a second radial direction **R2**, pointed towards the outlet end **162**, of the locking sleeve **108** is less than 90°. Preferably, the angle ranges from 30° to 60°. More preferably, the angle is 45°.

(53) When the second tubular body **204** of the male connector **200** is inserted between the locking sleeve **108** and the first tubular body **104** of the female connector **100** under the action of an external force, the locking protrusion **208** moves along the guiding groove **110** and applies a circumferential component force to the locking sleeve **108**, so that the locking sleeve **108** rotates. If the angle is too large, the locking protrusion **208** will be jammed when moving along the guiding

groove **110**, so that the locking sleeve **108** cannot be rotated in place, and thus the locking protrusion **208** cannot enter the locking groove **112** for final locking. If the angle is too small, the substantially triangular wall portion of the locking sleeve **108** between the guiding groove **110**, the locking groove **112** and the first end **150** will be relatively weak and cannot meet the pull-out resistance requirement of the connector assembly **10**. In this case, when the male connector **200** and the female connector **100** are locked to each other and the male connector **200** is subjected to an unexpected pulling force, the locking sleeve **108** may be damaged by the pulling force, which in turn causes the male connector **200** to be disconnected from the female connector **100**.

(54) Referring to FIG. **3A** again, during assembly of the female connector **100**, the locking sleeve **108** is sleeved outside the first tubular body **104** first, then the valve unit **106** is arranged inside the first tubular body **104**, and the fixing end **122** of the first tubular body **104** is inserted into the first receiving section **116** of the housing **102**, so that the shoulder **154** of the locking sleeve **108** is positioned between the first limiting portion **156** of the housing **102** and the second limiting portion **158** of the first tubular body **104**. Then, the fixing end **122** of the first tubular body **104** is fixed to the first receiving section **116** of the housing **102** by, for example, welding in the circumferential direction. So far, the assembly of the female connector **100** is finished. The above assembly process is only illustrative rather than restrictive.

(55) FIGS. **8A** to **13B** illustrate the male connector **200** of the connector assembly **10** according to the first embodiment of the present application and its various components.

(56) Referring to FIGS. **10A** to **10D**, the casing **202** of the male connector **200** may be substantially tubular, and include a second connection section **210**, a third receiving section **212** and a fourth receiving section **214** connected to each other in sequence. In the illustrated embodiment, the second connection section **210** may be in the form of a threaded connection section and may, for example, be fixed to a device using the connector assembly **10**. For example, the second connection section **210** may be threadedly engaged with a threaded hole (not shown) in a casing wall of a battery pack of a new energy vehicle, so as to be fixed to the battery pack. The casing **202** of the male connector **200** may be formed by injection molding.

(57) Referring to FIGS. **11A** and **11B**, the second tubular body **204** may include a second tubular main body **216** and the locking protrusion **208**. The second tubular main body **216** may be substantially tubular and include a fixing end **218** and a plug end **220**. The plug end **220** of the second tubular main body **216** defines an opening **222**. Optionally, the locking protrusion **208** may be made of metal, the second tubular main body **216** may be made of plastic, and the second tubular main body **216** and the locking protrusion **208** may be integrally formed by insert injection molding, so as to improve the pull-out resistance of the connector assembly **10**.

(58) In the illustrated embodiment, the second tubular body **204** may include two locking protrusions **208** oppositely arranged in its radial direction. Accordingly, the locking sleeve **108** of the female connector **100** may have two guiding grooves **110** and two locking grooves **112**.

(59) With reference to FIG. **9**, the second tubular body **204** of the male connector **200** is at least partially received within the fourth receiving section **214** of the casing **202**, and is fixed to the fourth receiving section **214** of the casing **202**.

(60) The fourth receiving section **214** of the casing **202** is configured to be capable of being mated with the fixing end **218** of the second tubular body **204**. Optionally, the fourth receiving section **214** may be in interference fit with the fixing end **218** of the second tubular body **204**, such that the fixing end **218** of the second tubular body **204** is fixed to the fourth receiving section **214**.

(61) Optionally, the fourth receiving section **214** and the fixing end **218** of the second tubular body **204** may be further fixed to each other by welding in the circumferential direction to provide higher fixing strength and achieve the sealing between the fourth receiving section **214** and the second tubular body **204**, thereby reducing the use of sealing members and saving costs.

(62) Optionally, as shown in FIG. **9**, the outer peripheral surface of the fixing end **218** of the second tubular body **204** may be provided with a second positioning protrusion **224**, and the

circumferential wall of the fourth receiving section **214** may be correspondingly provided with a second positioning recess **226**. The second positioning protrusion **224** and the second positioning recess **226** may be engaged with each other by snap fit, so as to improve the fixing strength of the fixing end **218** of the second tubular body **204** and the fourth receiving section **214**.

(63) It is conceivable that the casing **202** and the second tubular body **204** may be fixed to each other by interference fit, snap fit, welding or any combination thereof.

(64) In addition, referring to FIG. **9**, FIG. **10D** and FIG. **14C**, the third receiving section **212** of the casing **202** is configured to be capable of being mated with the first tubular body **104** of the female connector **100**. In other words, according to the actual application requirements, for example, in the case that the housing of the female connector **100** needs to have a threaded connection section, the casing **202** of the male connector **200** can also be used as the housing of the female connector **100** and be assembled and fixed with the first tubular body **104** of the female connector **100**, so that there is no need to manufacture a new mold for a housing having a threaded connection section for the female connector **100**.

(65) Thus, the third receiving section **212** and the fourth receiving section **214** of the casing **202** can respectively be mated with the first tubular body **104** of the female connector **100** and the second tubular body **204** of the male connector **200**, which can reduce the number of injection molds required for manufacturing the connector assembly **10** and significantly reduce manufacturing costs, as will be explained further below.

(66) The third receiving section **212** has a larger diameter than the second connection section **210**, and the fourth receiving section **214** has a larger diameter than the third receiving section **212**, so that a third inner stepped portion **228** is formed between the second connection section **210** and the third receiving section **212**, and a fourth inner stepped portion **230** is formed between the third receiving section **212** and the fourth receiving section **214**. The fourth inner stepped portion **230** is configured to limit the second tubular body **204** when the fourth receiving section **214** is mated with the second tubular body **204**.

(67) In addition, an outer stepped portion **232** (shown in FIG. **10D**) is further formed between the second connection section **210** and the third receiving section **212**. A sealing member **234** (as shown in FIG. **9**) may be embedded in the outer stepped portion **232** to achieve a sealing connection between the casing **202** and a casing wall of the battery pack of the new energy vehicle, for example.

(68) Referring to FIG. **9**, the valve assembly **206** of the male connector **200** includes a valve stem **236**, a sliding sleeve **238** and an elastic element **240**. The valve stem **236** includes at two ends thereof a valve stem head portion **241** and a valve stem base portion **242** respectively. The sliding sleeve **238** is arranged in the second tubular body **204** and is sleeved outside the valve stem **236**, and the sliding sleeve **238** is slidable between a second closed position and a second open position along the second axial direction **A2**. Two ends of the elastic element **240** respectively abut against the sliding sleeve **238** and the valve stem base portion **242** to bias the sliding sleeve **238** toward the second closed position. The sliding sleeve **238** blocks an annular gap between the second tubular body **204** and the valve stem head portion **241** to cut off a flow path of the male connector **200** when the sliding sleeve **238** is in the second closed position, and the flow path of the male connector **200** is opened when the sliding sleeve **238** is in the second open position. The valve stem **236** and the sliding sleeve **238** may be formed by injection molding.

(69) Referring to FIG. **9**, FIG. **12A** and FIG. **12B**, the valve stem **236** may be positioned in the second tubular body **204** in the second axial direction **A2**. The valve stem **236** may include the valve stem head portion **241**, the valve stem base portion **242**, and a valve stem middle portion **244** connecting the valve stem head portion **241** with the valve stem base portion **242**. In the illustrated embodiment, the valve stem **236** is integrally formed.

(70) The valve stem base portion **242** includes a base body **246** and a projection **248**. The base body **246** may be provided with a through hole **250** penetrating through the base body **246** for fluid

to flow therethrough. The projection **248** extends from the center of the base body **246** and away from the valve stem head portion **241** in the second axial direction **A2**. The projection **248** tapers in a direction away from the valve stem head portion **241**, so as to prevent the size of the cross section of the flow path of the male connector **200** from abruptly changing near the base body **246**, thereby avoiding turbulence. In the illustrated embodiment, the projection **248** is in the shape of a semi-ellipsoid. The valve stem middle portion **244** may be provided with multiple reinforcing ribs **252** to increase the strength of the valve stem **236**.

(71) Referring to FIG. **9**, FIG. **13A**, and FIG. **13B**, the sliding sleeve **238** is arranged in the second tubular body **204** and is sleeved outside the valve stem **236**. The sliding sleeve **238** is slidable between the second closed position and the second open position along the second axial direction **A2**. A sealing member **254** is provided between the outer peripheral surface of the sliding sleeve **238** and the inner peripheral surface of the second tubular body **204**. A sealing member **256** is provided between the inner peripheral surface of the sliding sleeve **238** and the outer peripheral surface of the valve stem head portion **241**.

(72) Two ends of the elastic element **240** abut against an inner stepped portion **258** of the sliding sleeve **238** and the valve stem base portion **242** respectively. The sliding sleeve **238** is biased towards the second closed position by the elastic force of the elastic element **240**. The outer periphery of the sliding sleeve **238** is provided with a limiting protrusion **260**. The inner circumference of the second tubular body **204** is provided with a limiting surface **262**. The limiting protrusion **260** and the limiting surface **262** may abut against each other to limit the sliding sleeve **238** in the second closed position where the sliding sleeve **238** is in sealing contact with the valve stem head portion **241**. The elastic element **240** may be in the form of a coil spring.

(73) When the sliding sleeve **238** is biased in the second closed position by the elastic element **240**, the sliding sleeve **238** cooperates with the sealing members **254** and **256** to block the annular gap between the second tubular body **204** and the valve stem head portion **241** to cut off the flow path of the male connector **200**. When the sliding sleeve **238** is pushed by an external force in the second axial direction **A2**, the sliding sleeve **238** can move away from the valve stem head portion **241** to the second open position against the elastic force of the elastic element **240**, so that the flow path of the male connector **200** is opened.

(74) As shown in FIG. **18C**, when the sliding sleeve **238** is in the second open position, fluid can enter the casing **202** through the gap between the sliding sleeve **238** and the valve stem **236**, then flow through the through hole **250** of the valve stem **236**, and then flow into the fluid pipeline (not shown) connected with the male connector **200**. And vice versa, fluid may flow along a reverse direction into the casing **202** from the fluid pipeline connected with the male connector **200** and then flow out through the gap between the sliding sleeve **238** and the valve stem **236**.

(75) Referring to FIG. **9** again, during assembly of the male connector **200**, the valve assembly **206** is arranged inside the second tubular body **204** first, and the fixing end **218** of the second tubular body **204** is inserted into the fourth receiving section **214** of the casing **202**. Then, the fixing end **218** of the second tubular body **204** is fixed to the fourth receiving section **214** of the casing **202** by, for example, welding in the circumferential direction. So far, the assembly of the male connector **200** is finished. The above assembly process is only illustrative rather than restrictive.

(76) Hereinafter, the process of connecting and locking the female connector **100** to the male connector **200** of the connector assembly **10** according to the first embodiment of the present application will be described with reference to FIGS. **14A** to **18C**.

(77) FIG. **14A** to FIG. **14C** show a state in which the female connector **100** and the male connector **200** are disconnected from each other. In this state, the valve core **136** of the female connector **100** is in the first closed position, and the sliding sleeve **238** of the male connector **200** is in the second closed position.

(78) During the connection of the female connector **100** and the male connector **200**, as shown in FIGS. **15A** to **15C**, the operator may first insert the second tubular body **204** of the male connector

200 between the first tubular body **104** and the locking sleeve **108** of the female connector **100**, so that the locking protrusion **208** of the second tubular body **204** is inserted into the inlet end **160** of the guiding groove **110** of the locking sleeve **108**. At this time, the valve stem **236** and the sliding sleeve **238** of the male connector **200** just come into contact with the valve core **136** and the plug end **124** of the first tubular body **104** of the female connector **100**, respectively.

(79) As the second tubular body **204** is further inserted, as shown in FIGS. **16A** to **16C**, the locking protrusion **208** moves along the guiding groove **110** and drives the locking sleeve **108** to rotate. During this process, the valve stem **236** of the male connector **200** pushes the valve core **136** from the first closed position toward the first open position against the elastic force of the elastic member **138** of the female connector **100**, and the plug end **124** of the female connector **100** pushes the sliding sleeve **238** from the second closed position toward the second open position against the elastic force of the elastic element **240** of the male connector **200**.

(80) Then, as shown in FIGS. **17A** to **17C**, the locking protrusion **208** enters the positioning end **164** of the locking groove **112**. Since the positioning end **164** of the locking groove **112** is located on one side of the outlet end **162** of the arc-shaped guiding groove **110** and there is a non-smooth transition between the outlet end **162** of the guiding groove **110** and the positioning end **164** of the locking groove **112**, the operator would get tactile feedback when the locking protrusion **208** enters the positioning end **164** of the locking groove **112**, and stop applying the insertion force to the male connector **200**.

(81) Then, as shown in FIGS. **18A** to **18C**, the female connector **100** and the male connector **200** tend to disengage/move away from each other under the action of the respective elastic forces of the elastic member **138** and the elastic element **240**, so that the locking protrusion **208** of the male connector **200** moves from the positioning end **164** of the locking groove **112** to the locking end **166** and engages with the locking end **166**, so as to prevent the second tubular body **204** of the male connector **200** from disengaging from the locking sleeve **108**, thereby locking the female connector **100** and the male connector **200** to each other. At this time, the valve core **136** of the female connector **100** is in the first open position, the sliding sleeve **238** of the male connector **200** is in the second open position, and the respective flow paths of the female connector **100** and the male connector **200** are opened and in fluid communication with each other.

(82) Arranging the locking sleeve **108** in a rotatable manner allows the female connector **100** to be connected and locked with the male connector **200** by means of straight insertion operation of the female connector and/or male connector during the connection of the female connector **100** and the male connector **200**. This straight insertion operation requires a smaller operation space and is convenient to operate, and is especially suitable for, for example, a battery pack with a small interior space of a new energy vehicle. And during the straight insertion operation, there is no need to rotate the housing **102** of the female connector **100**. This avoids the rotational resistance due to the fluid pipeline connected to the housing **102** of the female connector **100** when the housing **102** of the female connector **100** is rotated, and also reduces stress at the connection between the housing **102** of the female connector **100** and the fluid pipeline connected thereto.

(83) FIG. **19A** to FIG. **19C** illustrate a connector assembly **10A** according to a second embodiment of the present application. The connector assembly **10A** according to the second embodiment is substantially the same as the connector assembly **10** according to the first embodiment, and the only difference lies in that: the housing **102A** and the first tubular body **104A** of the female connector **100A** of the connector assembly **10A** according to the second embodiment are fixed to each other by interference fit and welding, and no snap fit structure for fixing the housing **102A** and the first tubular body **104A** is provided; and similarly, the casing **202A** and the second tubular body **204A** of the male connector **200A** are fixed to each other by interference fit and welding, and no snap fit structure for fixing the casing **202A** and the second tubular body **204A** is provided. The similarities between these two embodiments will not be repeated here.

(84) FIG. **20A** to FIG. **20C** illustrate a connector assembly **10B** according to a third embodiment of

the present application. The connector assembly **10B** according to the third embodiment is substantially the same as the connector assembly **10A** according to the second embodiment, and the only difference lies in that: the second connection section **210B** of the casing **202B** of the male connector **200B** of the connector assembly **10B** of the third embodiment is in the form of a hose connection section, and the casing **202B** of the male connector **200B** and the housing **102B** of the female connector **100B** have the same configuration. The similarities between these two embodiments will not be repeated here.

(85) FIG. **21A** to FIG. **21C** illustrate a connector assembly **10C** according to a fourth embodiment of the present application. The connector assembly **10C** according to the fourth embodiment is substantially the same as the connector assembly **10A** according to the second embodiment, and the only difference lies in that: the first connection section **114C** of the housing **102C** of the female connector **100C** of the connector assembly **10C** of the fourth embodiment is in the form of a threaded connection section, and the second connection section **210C** of the casing **202C** of the male connector **200C** is in the form of a hose connection section. The similarities between these two embodiments will not be repeated here.

(86) FIG. **22A** to FIG. **22C** illustrate a connector assembly **10D** according to a fifth embodiment of the present application. The connector assembly **10D** according to the fifth embodiment is substantially the same as the connector assembly **10A** according to the second embodiment, and the only difference lies in that: the first connection section **114D** of the housing **102D** of the female connector **100D** of the connector assembly **10D** of the fifth embodiment is in the form of a threaded connection section, and the housing **102D** of the female connector **100D** and the casing **202D** of the male connector **200D** have the same configuration. The similarities between these two embodiments will not be repeated here.

(87) Referring to FIGS. **19A** to **22C**, the connecting sections at two ends of the connector assemblies **10A**, **10B**, **10C**, and **10D** according to the second to fifth embodiments of the present application have different configurations, and can be adapted to different application scenarios.

(88) Components, except for the housing and the casing, of the connector assemblies **10A**, **10B**, **10C**, and **10D** according to the second to fifth embodiments of the present application have the same configuration. Therefore, when manufacturing the connector assemblies according to the second to fifth embodiments of the present application, plastic components of the connector assemblies other than the housing and the casing can share the injection molds.

(89) Besides, the housings and casings of the connector assemblies **10A**, **10B**, **10C**, and **10D** of the second to fifth embodiments may also share the injection molds. Specifically, taking the second embodiment as an example, similar to the first embodiment, referring to FIG. **19C**, the first receiving section **116A** and the second receiving section **118A** of the housing **102A** of the female connector **100A** of the connector assembly **10A** of the second embodiment are configured to be capable of being mated with the first tubular body **104A** of the female connector **100A** and the second tubular body **204A** of the male connector **200A**, respectively; and the third receiving section **212A** and the fourth receiving section **214A** of the casing **202A** of the male connector **200A** are configured to be capable of being mated with the first tubular body **104A** of the female connector **100A** and the second tubular body **204A** of the male connector **200A**, respectively. In other words, the housing **102A** of the connector assembly **10A** of the second embodiment can be mated with the first tubular body **104A** and/or the second tubular body **204A** according to actual needs, to be used as the housing of the female connector and/or the casing of the male connector. Similarly, the casing **202A** may also be used as the housing of the female connector and/or the casing of the male connector.

(90) Therefore, it is only necessary to customize the injection molds, for example, for the housing **102A** and the casing **202A** of the connector assembly **10A** of the second embodiment shown in FIGS. **19A** to **19C**, for example, to customize a first mold and a second mold for injection molding the housing **102A** and the casing **202A** respectively, which can meet the requirements for the

housing and the casing of the connector assemblies of the second to fifth embodiments. Specifically, both the housing **102B** and the casing **202B** of the connector assembly **10B** of the third embodiment may have the same configuration as the housing **102A** of the connector assembly **10A** of the second embodiment, and be manufactured with the first mold. The housing **102C** and the casing **202C** of the connector assembly **10C** of the fourth embodiment may respectively have the same configuration as the casing **202A** and the housing **102A** of the connector assembly **10A** of the second embodiment, and be manufactured with the second mold and the first mold respectively. Both the housing **102D** and the casing **202D** of the connector assembly **10D** of the fifth embodiment may have the same configuration as the casing **202A** of the connector assembly **10A** of the second embodiment, and be manufactured with the second mold.

(91) In this way, the number of custom molds required for manufacturing the connector assemblies of various configurations can be significantly reduced. And since the cost of preparing molds accounts for a considerable proportion of the manufacturing cost of plastic products, the manufacturing cost of the connector assemblies can be significantly reduced, while providing the connector assemblies with various configurations and expanding the application range of the connector assemblies.

(92) It should be understood that the embodiments shown in FIGS. **1** to **22C** only illustrate the shape, size and arrangement of each optional component of the connector assembly according to the present application. However, these embodiments are merely intended to illustrate, rather than limit. Other shapes, sizes and arrangements may be adopted without departing from the idea and scope of the present application.

(93) The technical contents and technical features of the present application have been disclosed above. However, it can be understood that, those skilled in the art can make various changes and improvements to the above-disclosed concept under the creative concept of the present application, and all these various changes and improvements still fall within the protection scope of the present application. The description of the foregoing embodiments is exemplary rather than restrictive, and the protection scope of the present application is determined by the appended claims.

Claims

1. A connector assembly comprising a female connector and a male connector for being connected with the female connector, wherein the female connector comprises: a housing; a first tubular body defining a first axial direction, the first tubular body being at least partially received within and fixed to the housing; a valve unit at least partially arranged in the first tubular body; and a locking sleeve sleeved outside the first tubular body in a rotatable manner, wherein at least part of the locking sleeve is confined between the housing and the first tubular body in the first axial direction, and the locking sleeve comprises a guiding groove and a locking groove which are connected to each other and arranged on a circumferential wall of the locking sleeve; the male connector comprises: a casing; a second tubular body defining a second axial direction, the second tubular body being at least partially received within and fixed to the casing, the second tubular body being configured to be insertable between the locking sleeve and the first tubular body in the first axial direction, the second tubular body comprising a locking protrusion arranged on an outer periphery thereof; and a valve assembly at least partially arranged in the second tubular body; wherein the locking protrusion is adapted to move along the guiding groove and drive the locking sleeve to rotate until the locking protrusion enters the locking groove during the insertion of the second tubular body between the locking sleeve and the first tubular body, and wherein the locking protrusion is adapted to engage with the locking groove due to interaction of the valve assembly and the valve unit after the locking protrusion enters the locking groove, to prevent the second tubular body from disengaging from the locking sleeve, wherein the locking sleeve has a first end and a second end, the guiding groove extends from the first end towards the second end, and the

guiding groove has an inlet end at the first end and an outlet end away from the first end, wherein the locking groove has a positioning end and a locking end opposite to each other, and the positioning end and the locking end are respectively located on two sides of the outlet end of the guiding groove, wherein the locking end is closer to the first end of the locking sleeve than the positioning end, and wherein there is a non-smooth transition between the outlet end of the guiding groove and the positioning end of the locking groove, wherein the locking protrusion is adapted to move along the guiding groove and enter the positioning end during the insertion of the second tubular body between the locking sleeve and the first tubular body, and the locking protrusion is further adapted to move from the positioning end to the locking end and engage with the locking end due to the interaction of the valve assembly and the valve unit after the locking protrusion enters the positioning end.

2. The connector assembly according to claim 1, wherein the housing has a first limiting portion, the first tubular body has a second limiting portion arranged on an outer periphery thereof, and the locking sleeve has a shoulder extending radially inward, and wherein the shoulder is confined between the first limiting portion and the second limiting portion.

3. The connector assembly according to claim 1, wherein the guiding groove is arc-shaped.

4. The connector assembly according to claim 3, wherein an angle between a first radial direction of the locking sleeve pointed towards the inlet end and a second radial direction of the locking sleeve pointed towards the outlet end is less than 90° .

5. The connector assembly according to claim 4, wherein the angle ranges from 30° to 60° .

6. The connector assembly according to claim 4, wherein the angle is 45° .

7. The connector assembly according to claim 1, wherein the locking groove extends in the first axial direction, and at least part of the locking groove extends from the outlet end of the guiding groove towards the first end of the locking sleeve.

8. The connector assembly according to claim 1, wherein the housing comprises a first connection section in a form of a hose connection section or a threaded connection section; and/or the casing comprises a second connection section in a form of a hose connection section or a threaded connection section.

9. The connector assembly according to claim 1, wherein the housing and the first tubular body are fixed to each other by interference fit, snap fit and/or welding; and/or the casing and the second tubular body are fixed to each other by interference fit, snap fit and/or welding.

10. The connector assembly according to claim 1, wherein the first tubular body has a plug end defining a port, wherein the valve unit comprises a valve core and an elastic member, the valve core is movable in the first axial direction between a first closed position and a first open position, and the valve core is biased toward the first closed position by the elastic member, wherein the valve core blocks the port to cut off a flow path of the female connector when the valve core is in the first closed position, and the flow path of the female connector is opened when the valve core is in the first open position.

11. The connector assembly according to claim 10, wherein the plug end defines a first inner peripheral inclined surface and a second inner peripheral inclined surface, wherein an inner peripheral surface of the port, the first inner peripheral inclined surface and the second inner peripheral inclined surface are sequentially connected in the first axial direction, and the first inner peripheral inclined surface and the second inner peripheral inclined surface are configured to guide a sealing member embedded in an outer periphery of the valve core to move in the first axial direction to come into sealing contact with the inner peripheral surface of the port, wherein the first inner peripheral inclined surface and the second inner peripheral inclined surface respectively form a first angle and a second angle with respect to the first axial direction, and the first angle is smaller than the second angle.

12. The connector assembly according to claim 11, wherein the first angle ranges from 5° to 15° .

13. The connector assembly according to claim 1, wherein the valve assembly comprises a valve

stem, a sliding sleeve and an elastic element, the valve stem comprises a valve stem head portion and a valve stem base portion at two ends of the valve stem respectively, the sliding sleeve is arranged in the second tubular body and sleeved outside the valve stem and is slidable between a second closed position and a second open position along the second axial direction, and two ends of the elastic element respectively abut against the sliding sleeve and the valve stem base portion to bias the sliding sleeve toward the second closed position, and wherein the sliding sleeve blocks an annular gap between the second tubular body and the valve stem head portion to cut off a flow path of the male connector when the sliding sleeve is in the second closed position, and the flow path of the male connector is opened when the sliding sleeve is in the second open position.

14. The connector assembly according to claim 13, wherein the valve stem base portion comprises: a base body provided with a through hole penetrating through the base body for fluid to flow therethrough; and a projection extending from a center of the base body and extending away from the valve stem head portion, the projection tapering in a direction away from the valve stem head portion.

15. The connector assembly according to claim 14, wherein the projection is in a shape of semi-ellipsoid.

16. The connector assembly according to claim 1, wherein the second tubular body comprises a second tubular main body, the locking protrusion is made of metal and the second tubular main body is made of plastic, and wherein the second tubular main body and the locking protrusion are integrally formed by insert injection molding.
